

one object, two exact faces, one form -- and one open frontier ($A \geq 0 \Leftrightarrow \text{RH}(L)$)

the icosian closure object

600-cell $\rightarrow$ maximal order $\mathcal{I}$

over $K = \mathbb{Q}(\sqrt{5})$, lattice E_8

GEOMETRY

E_8 root system

$\text{shell}_1 \cup (1/\varphi)\text{shell}_1 = 240$

EXACT

ARITHMETIC

$L(\Theta_I, s) = \zeta_K(s) \zeta_K(s-1)$

$C_2 = 1$, no local correction

EXACT

FRONTIER 

unified by the closure / trace form B (golden trace $c = 1 + 1/\sqrt{5} \leftrightarrow$ Weil form)